# Federated Adversarial Learning Using Blockchain for Intrusion Detection in Cyber Defense

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***Abstract*—Federated learning is new approach for cyber defense in which multiple clients are trained without sharing their raw data.This makes it acceptable for intrusion detection where security and privacy concerns are first priority.However different poisoning attacks like model poisoning can degrade the efficiency of system to high level.To tackle these types of challenges a blockchain techniques is implemented in federated learning architecture to maintain privacy. In the proposed approach, multiple clients are locally train on their data without sharing updates with server like centralized federated learning with integration of Blockchain to get untampered and accountable model to train next model with this data.Blockchain approach helps to reduce poisoning attacks and one point failure concerns. The purposed system is implemented on publicly available dataset NSL-KDD for intrusion detection and is evaluated on CPU and it gives 75 percent accuracy which demonstrate with availability of GPU it performs well and accuracy will touches 90 percent to detect intrusion and maintain privacy of data.Due to its decentralize nature there is no chance of single point of failure in this system.**

## I. Introduction

Recent advancement in technology field especially in artificial intelligence(AI) and machine learning(ML) drastically improves connectivity and efficiency in different domains.However these advancement also brought new threats and vulnerabilities regarding cyber attacks which are more challenging for the government,businesses and societies.An Intrusion Detection System(IDS) is most important component of a cyber security architecture that is develop to monitor and analyze network traffic for the indication of malicious activity and plays important role for the detection of threats and responds to these threats[1]. Traditional IDS architecture depends on centralized machine learning(ML) models in which raw model data were sent to central server for analysis.Although this architecture was effective in static networks but it also presents challenges in vehicular environments,including high latency, excessive bandwidth usage, privacy leakage, and limited scalability.In addition to this deep learning-based centralized IDSs mostly struggle to adapt to non-stationary, imbalanced, and dynamically changing vehicular data distributions[2,3].Federated learning(FL) model updates are shared to multiple server without sharing raw data due to this sensitivity and privacy of data is maintained.AI and ML first requires various dataset to train them successfully.In recent years, dataset size has been increased and models become more complex.Model training requires optimization and model's parameters distribution on multiple machines[4].The work purposed in this research is multi-layered architecture to provide security,robustness and privacy in IoT edge computing through integration of federated deep learning and blockchain architecture[5].

## II. Literature Review

Federated learning is a distributed machine learning technique that utilizes multiple servers to share model updates without exchanging raw data. This approach helps to address data sensitivity and protect privacy [4]. FL has shown exceptional capabilities in various fields, such as intelligent transportation, the Internet of Things (IoT), healthcare, manufacturing, agriculture, energy, remote sensing, and more. FL also transcends geographical limitations, facilitating efficient global collaboration [6]. As federated learning (FL) systems become increasingly personalized, they encounter new threats, including model poisoning and covert backdoor attacks. Promising defense strategies, such as adaptive layered trust aggregation and anomaly detection based on gradient similarity, can enhance robustness against these adversarial manipulations [7]. In federated learning (FL) systems, there are two communication methods: centralized design and decentralized design. In the centralized design, model parameters trained on clients are transmitted to the server. The updates are processed on the server and then sent back to the clients [8]. Communication between the server and clients can be either synchronous or asynchronous [8]. In a decentralized design, communication occurs between clients, and each client can update the central model parameters immediately [9]. Centralized design is commonly used. However, decentralized design is often preferred due to the potential risks associated with storing information on a single server. Nevertheless, designing a decentralized communication architecture presents significant challenges [10].

### A. Pros and Cons

Thanks to advancements in big data technologies and high-performance computing, machine learning (ML) applications have become increasingly relevant in data-driven fields like agri-technologies. Traditional ML relies on the computing resources and data available on a central server to maintain the relevance and consistency of its models. In these conventional scenarios, user data is stored centrally and used for testing and training to develop new and improved ML models. However, this centralized approach poses significant challenges, particularly concerning the security and privacy of users' personal information, as well as issues related to computational power and processing time [11]. Despite its advantages, federated learning has some significant drawbacks. Firstly, it does not effectively vet malicious participants, meaning that dishonest contributors can disrupt the model training process. Additionally, federated learning is susceptible to attacks from malicious servers, which can lead to serious threats, particularly through information leakage [12]. The centralized parameter server is at risk of being a single point of failure, which can affect all trainers. Additionally, there is a need to address the issues of inadequate incentives and the absence of a reliable reward distribution system for training federated learning models. Therefore, it is essential to establish an open and auditable system [13,14], decentralized [15], in centivized, and defensible [16] federated learning mechanism.

### B. Decentralized Federated Learning (DFL)

Decentralized Federated Learning (DFL) offers greater customization by eliminating the need for a central server. This approach can conserve both communication and computational resources while providing higher confidence across various scenarios. In DFL, the connections between peers in the communication network are adaptively configured and changed based on the specific situation. This flexibility is one of the key advantages of DFL. In addition to standard connection types such as line, ring, and fully connected configurations, it is possible to establish connections based on geographical proximity, client similarity, communication protocols, and other factors [9]. The main challenge of decentralized federated learning (FL) is building trust among participants, particularly when collaborating across different business organizations. In situations where mutual trust is lacking, the global model may not be reliable, and it becomes difficult to accurately measure the contributions of each participant [17]. Blockchain, a distributed ledger that originated from decentralized currency systems, provides distributed trust. This enables cooperation among participants without requiring mutual trust, by compelling them to act honestly [18]. Blockchain-distributed trust provides a novel approach to designing federated learning frameworks and paradigms. A key feature of blockchain is its capacity to allow untrusted participants to communicate securely with one another and exchange status update messages without depending on a fully trusted third party or an authorized central node. Consequently, blockchain can effectively tackle the challenges encountered in federated learning, leading to a methodology referred to as blockchain-based federated learning (BFL) [19]. Blockchain's central aggregator detects malicious and unreliable actors by automating smart contract execution to protect against federated learning poisoning attacks [20]. Sun et al. [21] developed a new blockchain platform to manage reputation values in a decentralized manner, ensuring accurate historical records of reputation, which significantly enhances the accuracy of Federated Learning (FL). Wang et al. [22] proposed a blockchain-based method for auditing encryption gradients that utilizes a behavior chain to log the encryption gradient from the data owner. This approach aims to enhance the security of federated learning (FL). Additionally, they developed a blockchain-assisted federated learning (BC-FL) framework to address the single point of failure associated with the central server in FL [19].

## III. Methodology

### A. Dataset

The NSL-KDD dataset was used for this research because of significantly improved class balancing and its wide use in intrusion detection research.NSL-KDD is updated version of KDD'99 and it uses less repeated instances and gives more realstic benchmark for machine learning-base IDS evaluation.

### B. Preprocessing

The dataset went through number of preprocessing steps to validate high level learning on the basis of quality.Categorical features were converted into numerical features by using One-Hot Encoding.By using StandardScaler, we normalize numerical features to uniform scale across entire data inputs.To deal with class imbalancing in data we use the Synthetic Minority Oversampling Technique (SMOTE).After preprocessing we find 1,376,254 samples with 140 features.Preprocessing was essential to prevent model form bias ness and to achieve stable federated learning(FL).

### C. Federated Learning Setup

Five clients were used in federated learning(FL) setup.The data set was distributed equally through each client to obtain distributed data environments.Each client was trained on its received data for five epochs.After local training,model updates were sent to central server for the aggregation using Federated Aver aging (FedAvg) algorithm.This method preserves the privacy of the data and prevents data leakage.

### D. Adversarial Learning

FGSM-based perturbations were used to increase the robustness against adversarial threats in local training phase.Each client was trained on small noise to resist such type of threats.This type of training ensure that model is capable to resist these types adversarial threats and perform well when they are deployed in real-world applications.

### E. Blockchain Integration

Blockchain technique was used to secure federated learning process.For each global training round a block was generated containing the round number, SHA 256 hash of the aggregated model, and the previous block's hash. This created a sequential, immutable chain of training events.At the end of training process suspicious or malware modifications in the model updates were discarded and it is verified by the "Blockchain verified = True.

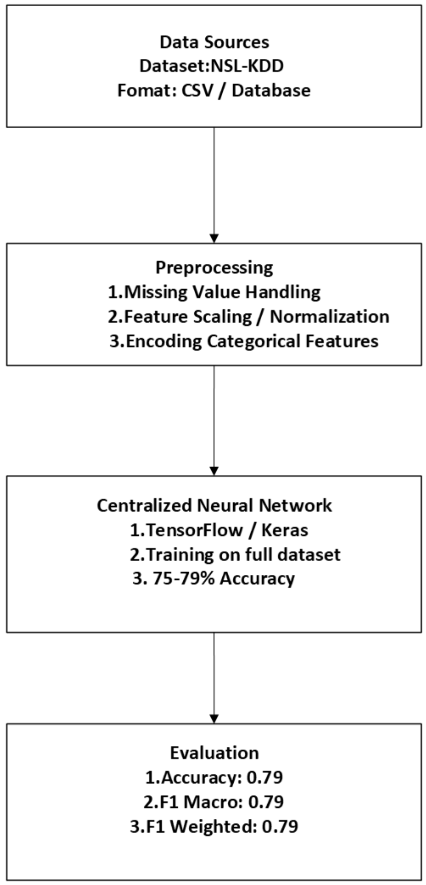


Fig. 1: Baseline Model Architecture

## IV. Implementation Environment

The entire implementation was done on the google colab by using CPU only.TensorFlow 2.19.0 was used for deep learning process,scikit-learn 1.6.1 used for preprocessing and evaluation utilities, and imbalanced-learn 0.14.0 are used for SMOTE-based resampling.Although in absence of GPU this federated leaning(FL) framework successfully completed three rounds in reasonable time period for these computation process.

### A. *Training Summary*

The federated adversarial learning process was executed for three global rounds. All five clients completed their local training tasks during each round without interruption. After every round, the aggregated model was hashed and appended to the blockchain, maintaining a complete and verifiable record of updates. No block inconsistencies or integrity violations were observed throughout the training.

### B. *Final Accuracy*

The final multiclass accuracy was 75.14 percent.This accuracy demonstrates viability for this process of combing federated learning, blockchain, and adversarial training—even in a CPU-only environment.

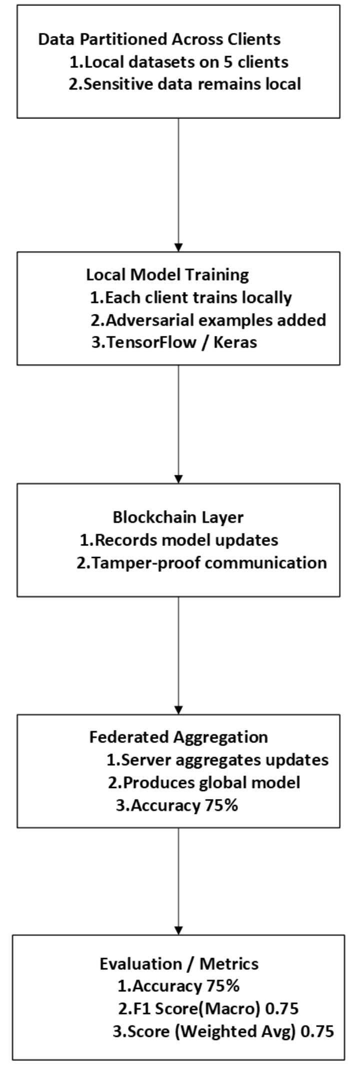


Fig. 2: Proposed Federated Adversarial Learning Architecture

### C. *Blockchain Verification*

On completing training phase blockchain verifies that all blocks,hashes and references were intact.No unauthorized modifications were done on the model during training process.

TABLE I: Comparison of Baseline and Federated Models

| Metric | Baseline Model | Federated Model |
|---|---|---|
| Accuracy | 79.02% | 75.13% |
| F1 Macro | 0.7903 | 0.7499 |
| F1 Weighted | 0.7902 | 0.7498 |
| Precision (avg) | 0.77 | 0.77 |
| Recall (avg) | 0.75 | 0.75 |

### D. *Observations*

Although accuracy was lowered but it can be improved by the use of GPU by increasing epochs and frame size.But this architecture ensures that it can offer privacy,sustainability and robustness for the cyber defense environments.Blockchain provides the accountability of models to ensure poison-free model for global model training.

## V. Discussion

The combination of federated leaning,adversarial techniques and blockchain provides strong defense system against threats.This architecture provides privacy and robustness.Due to its decentralized nature we can overcome single point failure problem.By the use of blockchain we can get rid of model poisoness problem to ensure the model to train on authentic data.Although accuracy is lowered in this process but it can be improved by use of GPU by changing different factors like epochs and frames during training process.

## VI. Conclusion

This research successfully demonstrates the implementation of a blockchain enhanced Federated Adversarial Learning framework for intrusion detection in cyber defense environments. The system achieved a multiclass accuracy of 75.14 percent using CPU-based training, validating its practical feasibility. The integration of adversarial learning improves resilience against malicious perturbations, while blockchain ensures tamper-proof and transparent model aggregation. The proposed approach shows strong potential for developing secure, decentralized, and robust IDS solutions. Future work may incorporate GAN based defenses, differential privacy, and deployment in real-world distributed networks. Future improvements include: Although the proposed blockchain secured Federated Adversarial Learning framework demonstrates promising performance, several enhancements can further strengthen its effectiveness in real world cyber defense environments. Future work may incorporate Generative Adversarial Networks (GANs) to create more sophisticated adversarial samples, enabling stronger model robustness against advanced evasion and poisoning at tacks. Differential Privacy (DP) can also be integrated to provide formal privacy guarantees and protect sensitive client information during model updates. Additionally, implementing secure aggregation and homomorphic encryption would further enhance confidentiality within the federated communication process. Deploying the system across real-world distributed network environments—such as IoT devices, cloud infrastructures, and enterprise networks—would allow evaluation under practical constraints such as communication delays, heterogeneous hardware, and varying data distributions. Finally, expanding blockchain capabilities with smart contracts could automate client verification, update validation, and anomaly detection, making the system more autonomous, scalable,and secure.

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